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CHUCK YEAGER

ON EXCEEDING THE SPEED OF SOUND



HEADING FOR THE BARRIER

Photographed from a chaser plane, the rocket-powered Bell X-1 accelerates toward supersonic speed. The experimental flight pictured here took place in 1951, after Glamorous Glennis had been repainted white instead of its original orange.

by loss of control over the elevator – the very phenomenon that had led many people to believe that aircraft would never fly beyond the speed of sound. But Yeager was armed with a technique for avoiding loss of control, by shifting from controlling pitch with the elevator to manipulating the movable stabilizer on the tail.

Yeager rocketed on up to 42,000ft (13,000m), still gathering speed. “I noticed that the faster I got, the smoother the ride,” he wrote. Then came the great moment. “Suddenly the Mach needle began to fluctuate. It went up to .965 Mach – then tipped right off the scale. I thought I was seeing things! We were flying supersonic! And it was as smooth as a baby’s bottom: Grandma could be sitting up there sipping lemonade.”

When Yeager glided down to land at Muroc, there was no hero’s welcome waiting. The secrecy

surrounding the X-1 program extended even to a ban on too flamboyant a private celebration. A few beers were drunk and a few steaks were eaten, and that was all. But the crucial point had been made. As Yeager later wrote: “The real barrier wasn’t in the sky, but in our knowledge and experience of supersonic flight.”

MACHOMETER

The X-1’s instrument panel featured a machometer – Mach 1 being the speed of sound. The one used during the first supersonic flight measured Mach numbers only up to Mach 1. This machometer was installed for subsequent flights.



There was widespread speculation that a “sound barrier” might block the path to any further rise in airspeed. Since no one had ever flown faster than sound, this was at least plausible. Pilots who had reached high subsonic speeds flying late-World War II piston-engined aircraft in steep dives reported violent turbulence. Another disturbing experience approaching the speed of sound was sudden loss of control of the aircraft – the controls seemed to freeze, as if all the cables to the control surfaces had been cut.

The sound barrier

High-speed flight has since become so commonplace, that it is difficult to grasp the anxieties that surrounded rising velocities in the late 1940s. No human being had ever traveled at anything approaching the speed of these jet aircraft. It was natural to ask whether there was a limit to what the human frame could stand, or a limit to the speed at which an aircraft could fly.

Speculation about the sound barrier was heightened by unexplained crashes. Chief MiG test pilot Alexei Grinchik was killed in May 1946, when the MiG-9 prototype he was flying went out of control. This concerned Soviet researchers but was unknown to the outside world. However, when Geoffrey de Havilland, son of the founder of the eponymous British aircraft company and its chief test pilot, was killed in a crash in September 1946 it caused a sensation. De Havilland was flying the D.H.108, an experimental, tailless swept-wing aircraft with the fuselage of a Vampire, when the aircraft broke up in flight.

The popular press labeled him immediately a victim of the invisible barrier in the sky.

The “right stuff”

The mystique of the sound barrier gave a special edge to experiments with the air-launched Bell X-1 rocket aircraft that began in 1946 at Muroc (later Edwards Air Force Base) in the California desert. The X-1 was designed purely as a test vehicle to allow NACA scientists to monitor the effects of high-speed flight. The 18 test pilots chosen for the program included Bell’s Chalmers “Slick” Goodlin and the USAAF’s Captain Charles “Chuck” Yeager. It was the military flier who had the temperament to “push the outside of the envelope.”

There were plenty of scares as, in flight after flight, Yeager pushed the X-1 toward the sound barrier. At .86 Mach, violent buffeting set in, like “driving on bad shock absorbers over uneven